

Candidate's Declaration

I, Md. Marzan Haque, hereby declare that the project work entitled "BookCycleBD - An Online Second-Hand Book Marketplace" submitted to the Department of Computer Science and Engineering, Rajshahi Science and Technology University (RSTU), Natore, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering, is an original work carried out by me under the supervision of Md. Shahidul Islam Shabuz, Assistant Professor, Department of Computer Science and Engineering, RSTU, Natore.

I further declare that this project report or any part of it has not been submitted to any other institution or university for the award of any degree or diploma.

Md. Marzan Haque

Registration No.: 0732220005101010

Department of Computer Science and Engineering

Rajshahi Science and Technology University (RSTU), Natore

Date: May 23, 2026

Certificate

This is to certify that the project report entitled "BookCycleBD - An Online Second-Hand Book Marketplace" submitted by Md. Marzan Haque, Registration No. 0732220005101010, has been carried out under my supervision in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

To the best of my knowledge, the work presented in this report is original and suitable for submission.

Supervisor

.....

Md. Shahidul Islam Shabuz

Assistant Professor

Department of Computer Science and Engineering

Rajshahi Science and Technology University (RSTU), Natore

Letter of Submission

23 May 2026

To

Dewan Juel Rahman

Associate Professor and Head

Department of Computer Science and Engineering

Rajshahi Science and Technology University (RSTU), Natore

Subject: Submission of Final Year Project Report

Dear Sir,

With due respect and humble submission, I beg to state that I have completed my final year project entitled "BookCycleBD - An Online Second-Hand Book Marketplace" under the supervision of Md. Shahidul Islam Shabuz, Assistant Professor, Department of Computer Science and Engineering, Rajshahi Science and Technology University (RSTU), Natore.

I have chosen this topic because affordable access to books remains a critical challenge for students in Bangladesh. A dedicated peer-to-peer book exchange platform can make a meaningful social and economic impact as the country continues to advance in digital services.

I hereby submit the project report for your kind consideration and evaluation as part of the requirements for the degree of Bachelor of Science in Computer Science and Engineering. I have devoted my utmost sincerity and dedication to making this report accurate, systematic, and reliable.

I remain available to provide any further clarification regarding this project at your convenience.

Yours sincerely,

Md. Marzan Haque

Registration No.: 0732220005101010

Department of Computer Science and Engineering

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Acknowledgement

First and foremost, I express my sincere gratitude to the Almighty Allah for granting me the ability, patience, and determination to complete this project successfully.

I would like to express my heartfelt gratitude to my respected supervisor, Md. Shahidul Islam Shabuz, Assistant Professor, Department of Computer Science and Engineering, Rajshahi Science and Technology University (RSTU), Natore, for his invaluable guidance, continuous encouragement, and constructive feedback throughout the entire project development process.

I am also grateful to the Head of the Department and all faculty members of the Department of Computer Science and Engineering, RSTU, for their academic support and encouragement during the course of my undergraduate studies.

I acknowledge the support of my family, especially my parents, whose unwavering encouragement and sacrifices have been the foundation of my academic achievements.

I also acknowledge the assistance received from various online documentation resources, open-source communities, and fellow students whose work and discussions contributed to the development of this project.

Finally, I am grateful to Rajshahi Science and Technology University for providing the opportunity to undertake this final year project as part of the Bachelor of Science in Computer Science and Engineering programme.

Abstract

BookCycleBD is a web-based peer-to-peer marketplace designed for buying and selling second-hand books in Bangladesh. The project addresses the problem of expensive academic books and the absence of a dedicated digital platform for structured book exchange. Existing general classified websites and social media groups do not provide book-specific features such as filtering by author, ISBN, category, condition, and location.

The main objective of this project is to develop a user-friendly platform where users can register, post book advertisements, upload book images, search and filter listings, communicate with sellers, and manage their own advertisements. The system also includes an administrative panel for managing users, advertisements, categories, and user reports.

The application was developed using Next.js, React, TypeScript, Supabase PostgreSQL, NextAuth.js, Cloudinary, Tailwind CSS, and Vercel. The project follows the software development lifecycle, including requirement analysis, system design, database design, implementation, testing, and deployment.

The final system provides a practical solution for students and readers by promoting affordable book access and encouraging book reuse.

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Chapter 1: Introduction

1.1 Introduction

The book industry in Bangladesh, particularly the academic book sector, faces a persistent challenge: new books are expensive and often beyond the reach of many students. At the same time, students who have finished using their books have no reliable platform to sell them to those who need them. General-purpose classified platforms such as Bikroy.com exist, but they are not optimized for books — there is no filtering by author, ISBN, condition, or academic subject.

BookCycleBD is developed to address these challenges by providing a dedicated, community-driven online marketplace exclusively for second-hand books in Bangladesh. The platform enables peer-to-peer book exchange through classified advertisements. Users can post book listings, browse and search by various criteria, and negotiate directly through a built-in messaging system. A full administrative panel ensures content quality and platform integrity.

1.2 Problem Statement

Students in Bangladesh often face financial difficulty in purchasing new academic books. At the same time, many students have used books that remain unused after completing a semester or course. Existing platforms such as general classified websites and social media groups are not optimized for book exchange because they lack structured book listings, book-specific filters, organized communication, and proper moderation. Therefore, a dedicated online platform is needed where users can easily post, search, buy, and sell second-hand books in a structured and reliable way.

1.3 Objectives

The main objectives of BookCycleBD are:

1. To develop a web-based platform for buying and selling second-hand books.
2. To allow registered users to post book advertisements with title, author, ISBN, category, condition, price, location, description, and images.
3. To provide search and filtering features based on keyword, category, condition, price range, and location.

4. To enable direct communication between buyers and sellers through an internal messaging system.
5. To implement role-based authentication and authorization for users and administrators.
6. To provide an admin panel for managing users, advertisements, categories, and reports.
7. To design a responsive user interface suitable for desktop, tablet, and mobile devices.
8. To promote book reuse and reduce the financial burden of purchasing new books.

1.4 Motivation

The motivation behind BookCycleBD stems from a real and everyday problem observed among students in Bangladesh. Academic textbooks are costly, and students who have finished using them rarely have a convenient way to pass them on to others who need them. Existing general-purpose classified platforms lack the book-specific features needed to make this exchange efficient and trustworthy.

The project aspires to build a focused, niche marketplace that empowers students and book lovers to participate in a circular economy for books — reducing waste, reducing cost, and making knowledge more accessible. The growing digital infrastructure of Bangladesh makes this the right time for such a platform.

1.5 Scope of the Project

The scope of BookCycleBD is limited to the development of a web-based second-hand book marketplace. The system allows users to register, log in, post book advertisements, upload images, search and filter books, communicate with sellers, and manage their own listings. The system also provides an admin panel for managing users, advertisements, categories, and reports. The current version does not include online payment, delivery tracking, seller ratings, or a dedicated mobile application. These features may be added in future versions.

1.6 Expected Outcome

BookCycleBD is expected to deliver a responsive and user-friendly web application accessible across all devices. The platform will provide:

- A seamless advertisement posting experience with image upload and category selection.
- Efficient book discovery through search, filter, and browse features.

- Direct buyer-seller communication through an integrated messaging system.
- A secure, role-based access system with distinct privileges for users and administrators.
- A live, production-deployed platform accessible to students across Bangladesh.

1.7 Report Organization

This report is organized into eight chapters. Chapter 1 provides an introduction, problem statement, objectives, motivation, and scope. Chapter 2 reviews existing systems and their limitations. Chapter 3 describes the development methodology and requirement analysis. Chapter 4 presents the system design including architecture, use case diagrams, and ER diagram. Chapter 5 explains the implementation details. Chapter 6 presents the user interface designs. Chapter 7 covers testing and evaluation. Chapter 8 provides the conclusion, limitations, and future scope.

Chapter 2: Literature Review and Existing System Analysis

2.1 Existing Systems

Several platforms are currently used in Bangladesh for buying and selling second-hand books. However, none of them provides a dedicated, structured, and feature-rich solution for book exchange. The major existing alternatives are outlined below.

Existing System	Limitations
Facebook Groups	Unstructured posts, poor search, no proper filtering, unreliable communication
Bikroy.com	General marketplace, not optimized for books, limited book-specific fields
Physical Bookshops	Location dependent, limited stock, requires physical visit
Personal Contacts	Limited reach and informal transaction process
BookCycleBD (Proposed)	Dedicated book platform with structured listing, filtering, messaging, and admin moderation

2.2 Limitations of Existing Systems

Existing platforms have several key limitations. General classified sites like Bikroy.com lack book-specific search filters such as ISBN, author, academic subject, and book condition. Social media groups used for book exchange have no structured listing format, no search functionality, and no secure messaging. No dedicated peer-to-peer book marketplace exists in Bangladesh that integrates advertisement management, messaging, and admin moderation in a single platform.

2.3 Proposed System

BookCycleBD overcomes these limitations by providing a dedicated platform for second-hand book exchange. It supports structured book advertisements, image upload, category-based browsing, search and filtering, buyer-seller messaging, and administrative moderation. As a result, the proposed system is more suitable for students and readers than general-purpose classified platforms.

2.4 Comparative Analysis

The following table presents a comparative analysis of existing systems and BookCycleBD across key features relevant to a book marketplace:

Feature	Facebook Groups	Bikroy.com	Physical Shops	BookCycleBD
Book-specific search filters	No	Partial	No	Yes
Image upload	Yes	Yes	No	Yes
Buyer-seller messaging	Partial	No	In-person	Yes
Admin moderation	No	Partial	N/A	Yes
ISBN / Author filter	No	No	No	Yes
Condition / Price filter	No	Partial	No	Yes
Accessible on mobile	Yes	Yes	No	Yes

Chapter 3: Methodology and System Analysis

3.1 Development Methodology

This project followed the Agile software development methodology. Agile was selected because it supports iterative development, continuous testing, and flexible modification of requirements. The system was divided into several modules, including authentication, advertisement management, search and filtering, messaging, user profile management, and admin management.

Each module was designed, implemented, tested, and improved step by step. This approach helped to identify errors early and allowed the project to be updated according to functional requirements.

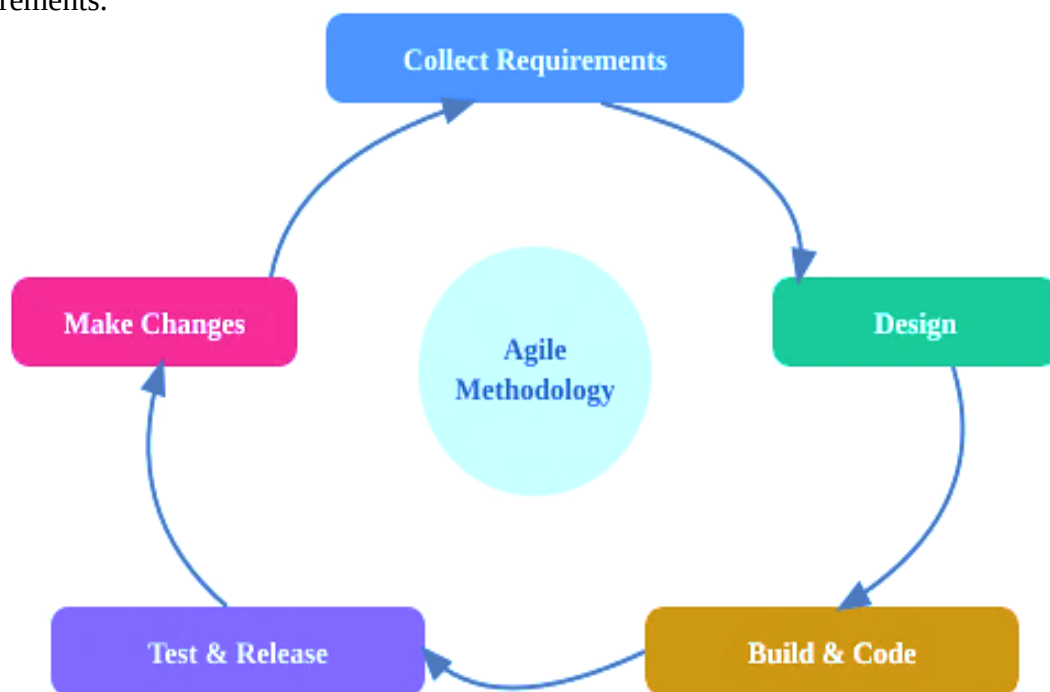


Figure 1: Agile Methodology Loop

Iteration	Module Developed	Output
Iteration 1	Requirement analysis	Functional and non-functional requirements
Iteration 2	Authentication module	Registration and login system
Iteration 3	Advertisement module	Book posting, editing, deleting
Iteration 4	Search and filter module	Book discovery features
Iteration 5	Messaging module	Buyer-seller communication
Iteration 6	Admin module	User, advertisement, category, and report management
Iteration 7	Testing and deployment	Final deployed application

3.2 Requirement Analysis

Requirement analysis identifies what the system should do and what constraints it must satisfy. The requirements of BookCycleBD are divided into functional requirements, non-functional requirements, and development requirements.

3.2.1 Functional Requirements

1. Users can register and log in.
2. Users can create, edit, and delete book advertisements.
3. Users can upload one or more book images.
4. Users can search and filter books by title, author, category, price, condition, and location.
5. Users can contact sellers through the messaging system.
6. Sellers can manage their own advertisements.
7. Admins can manage users, advertisements, categories, and reports.
8. The system can restrict unauthorized access to protected pages.

3.2.2 Non-Functional Requirements

1. Performance: The system should load pages quickly and respond efficiently to user actions.
2. Security: Passwords must be securely hashed, and protected routes must be accessible only to authorized users.
3. Usability: The interface should be simple and understandable for general users.
4. Responsiveness: The system should work properly on mobile, tablet, and desktop devices.
5. Maintainability: The codebase should be modular and easy to update.
6. Scalability: The database and image storage system should support future growth.

3.2.3 Development Requirements

Node.js, Next.js, React, TypeScript, Supabase PostgreSQL, NextAuth.js, Cloudinary, Tailwind CSS, VS Code, and Vercel.

3.3 Feasibility Study

3.3.1 Technical Feasibility:

The project is technically feasible because it uses widely available web technologies such as Next.js, React, Supabase, Cloudinary, and Vercel. These tools support rapid development, cloud database management, authentication, image storage, and deployment.

3.3.2 Operational Feasibility:

The system is easy to use for students, readers, and administrators. Users can post and search book advertisements without advanced technical knowledge.

3.3.3 Economic Feasibility:

The project can be developed with free or low-cost cloud services. Therefore, it is economically feasible for an undergraduate project.

3.3.4 Schedule Feasibility:

The project can be completed within the academic project timeline by dividing development into modules such as authentication, advertisement management, search, messaging, admin panel, testing, and deployment.

Chapter 4: System Design

4.1 System Architecture

BookCycleBD follows a client-server architecture. The client side is accessed through a web browser, where users interact with the interface. The application is developed using Next.js, which handles both frontend rendering and backend API routes. When a user sends a request, the Next.js server processes the request and communicates with the Supabase PostgreSQL database to retrieve or store data. Authentication is handled using NextAuth.js, which manages secure user sessions. Book images are uploaded to Cloudinary, and the stored image URLs are saved in the database. The application is deployed on Vercel, allowing users to access the system online. This architecture improves maintainability, scalability, and performance by separating user interaction, business logic, database storage, and media management.

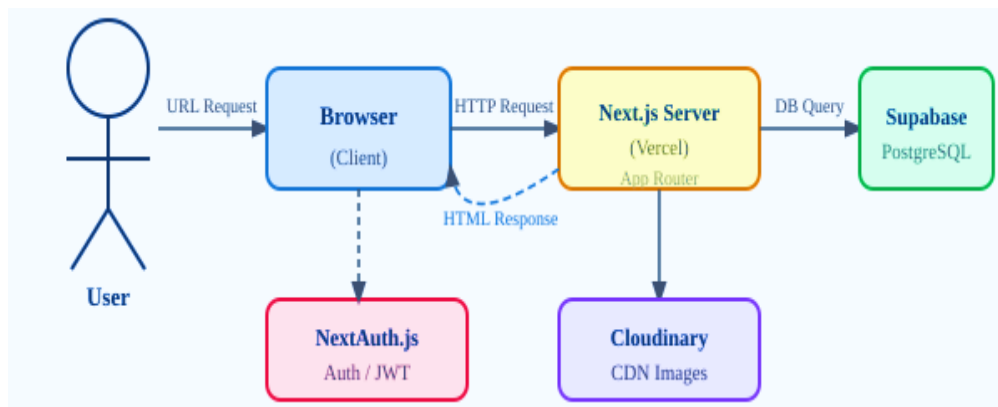


Figure 2.1: Architecture Design

4.2 Use Case Diagram

A use case diagram is a graphical representation of the interactions between users (actors) and the system. It depicts what the system does from the perspective of the user and makes it easier to communicate between the client and the system developers.

The main actors of the system are Buyer, Seller, Admin, and Guest User. A guest user can browse books but must register or log in to post advertisements or contact sellers. A buyer can search books, view advertisement details, contact sellers, send messages, and manage profile information. A seller can post advertisements, edit listings, delete listings, mark books as sold, and reply to buyer messages. The admin can manage users, advertisements, categories, and reports.

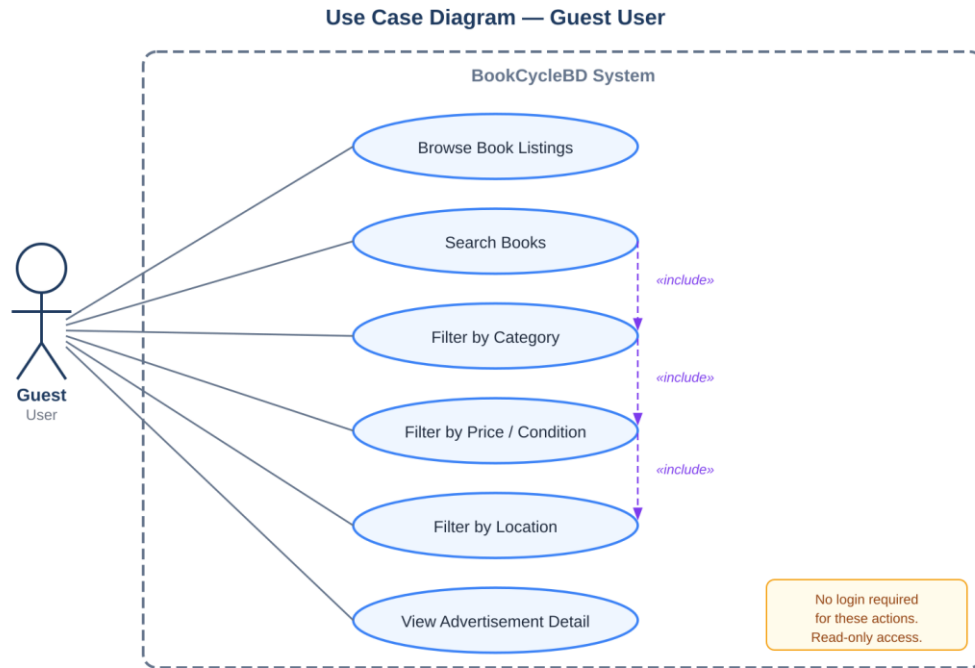


Figure 2.2: Use Case Diagram for guest user

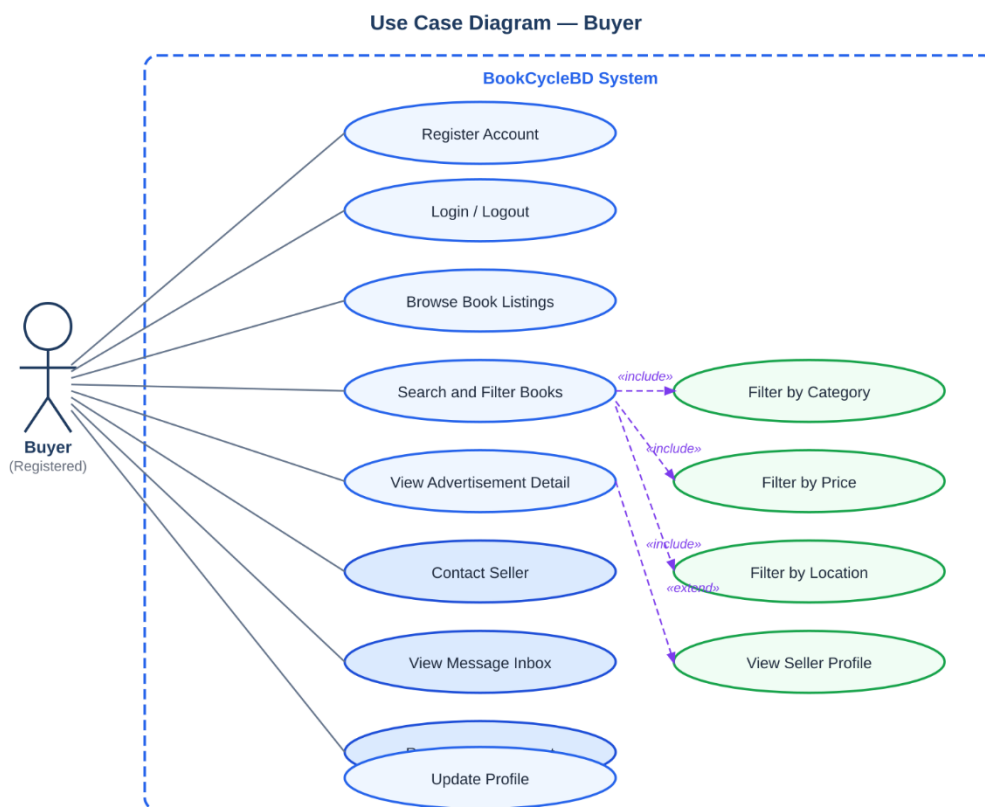


Figure 2.2: Use Case Diagram for Buyer

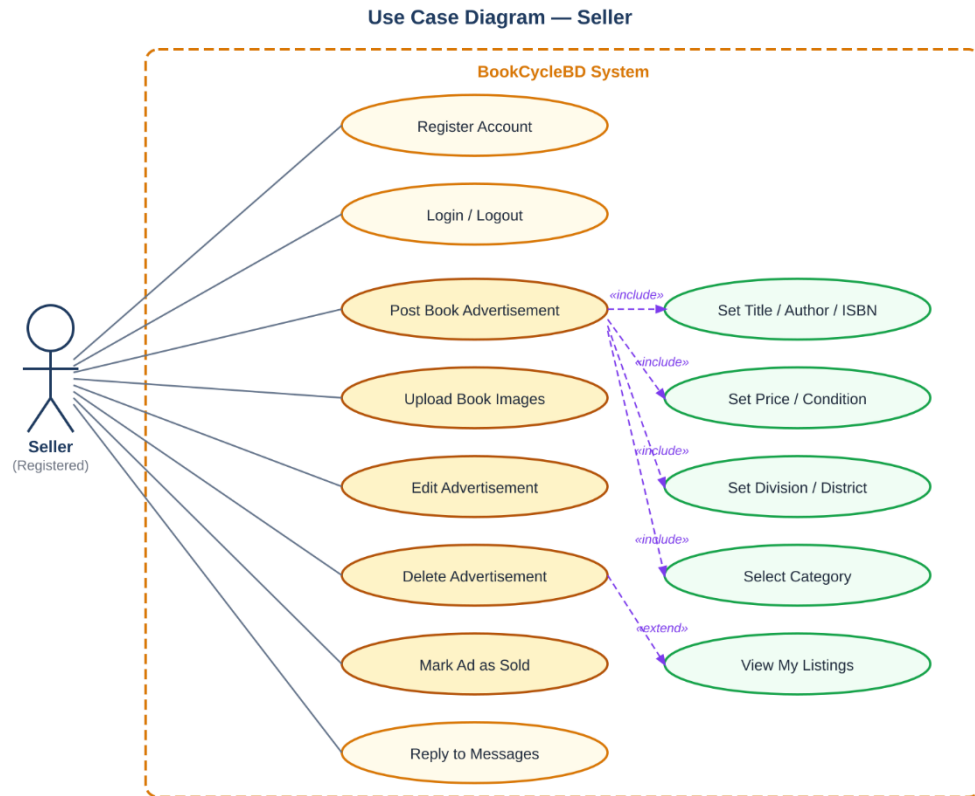


Figure 2.3: Use Case Diagram for Seller

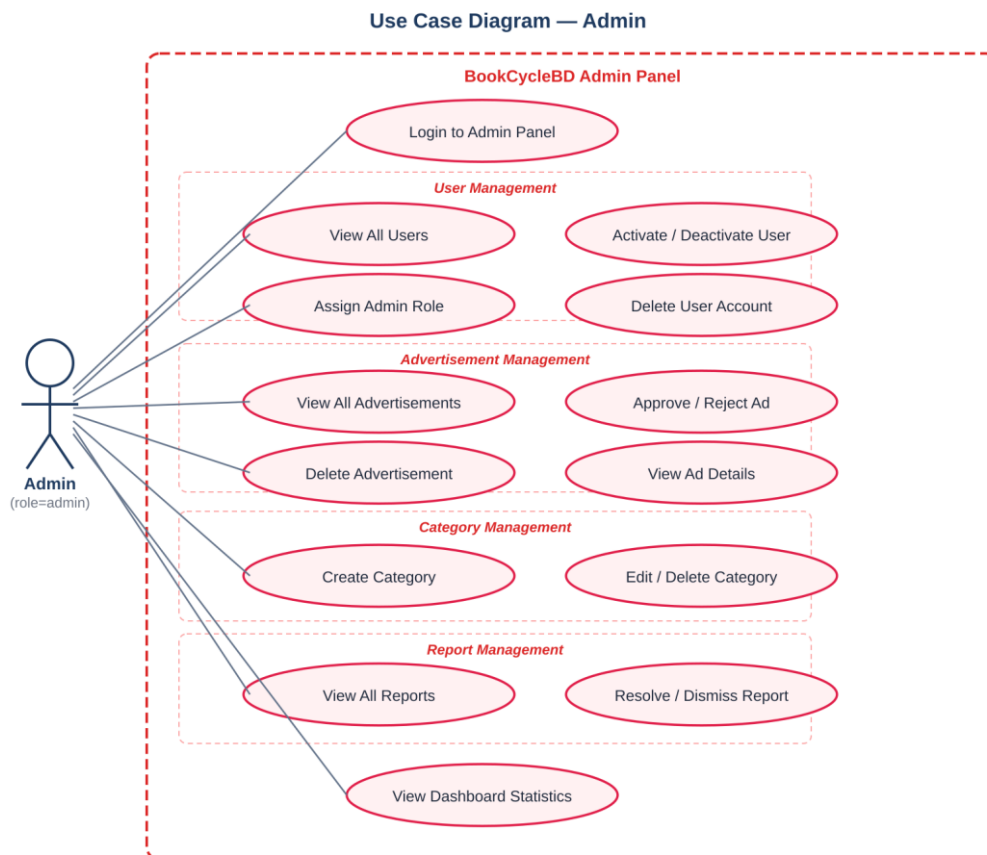


Figure 2.4: Use Case Diagram for Admin

4.3 ER Diagram

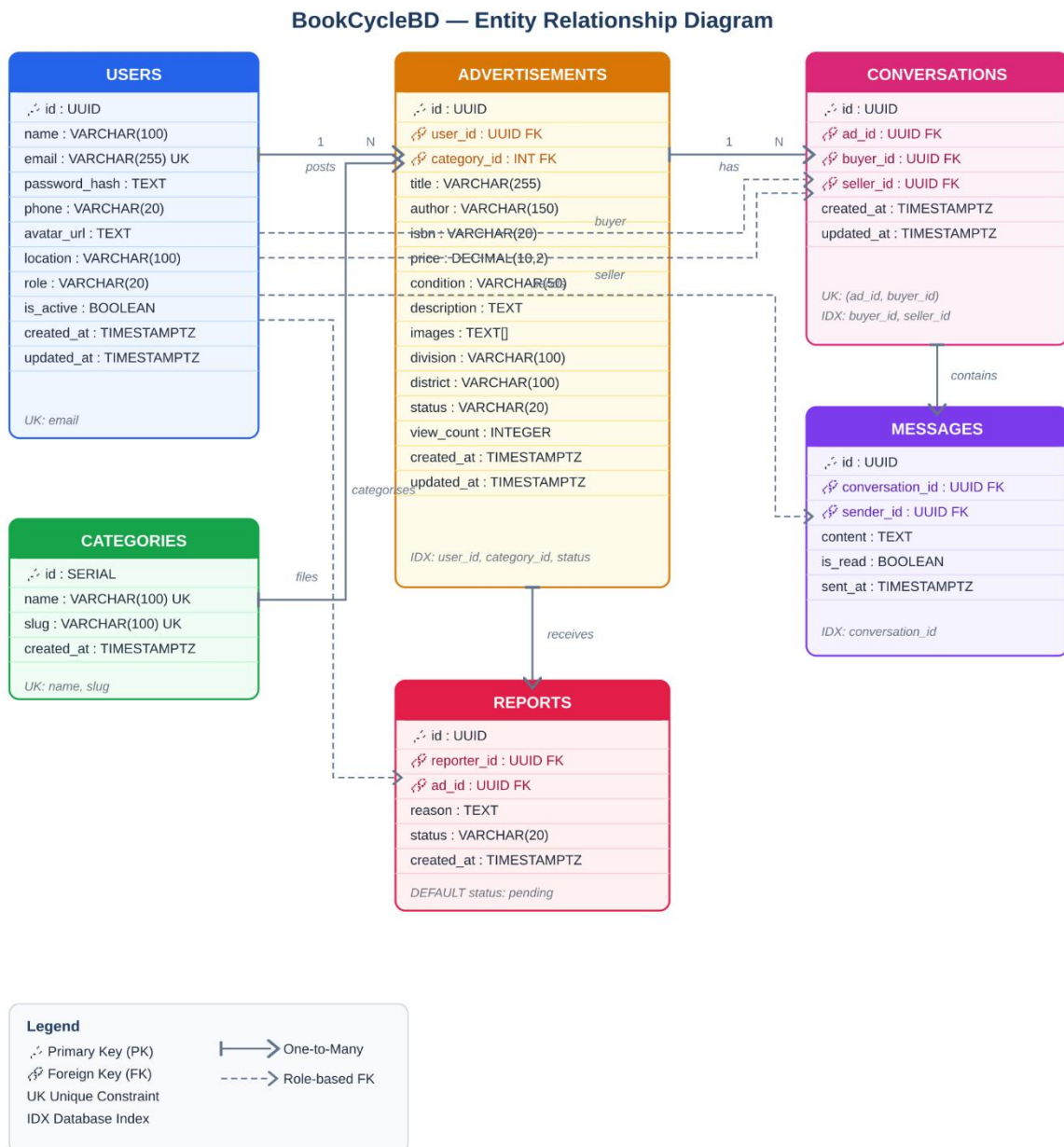


Figure 2.5: ER Diagram

The database of BookCycleBD consists of six major entities: users, advertisements, categories, conversations, messages, and reports. A user can post multiple advertisements, but each advertisement belongs to only one user. Each advertisement is assigned to one category, while a category can contain many advertisements. A conversation is created between a buyer and a seller for a specific advertisement. Each conversation can contain multiple messages. The reports table stores user-submitted reports against advertisements, which can be reviewed by administrators. These relationships help maintain data consistency and support the major features of the system.

4.4 Database Design

The following table describes the major database tables used in BookCycleBD:

Table Name	Purpose
users	Stores user account and profile information
advertisements	Stores book advertisement details
categories	Stores book categories
conversations	Stores buyer-seller conversation records
messages	Stores individual messages
reports	Stores user reports against advertisements

Chapter 5: Implementation

5.1 Technology Stack

The system was developed using Next.js, React, TypeScript, Supabase PostgreSQL, NextAuth.js, Cloudinary, Tailwind CSS, react-hook-form, bcryptjs, and Vercel. These technologies were selected because they support modern full-stack web development, secure authentication, cloud database management, image hosting, responsive design, and easy deployment.

Technology	Purpose
Next.js	Full-stack framework with App Router, SSR, and API routes
React	Component-based UI library
TypeScript	Strongly typed JavaScript for improved reliability
Supabase (PostgreSQL)	Cloud-hosted relational database
NextAuth.js	Authentication with JWT session management
Cloudinary	Image upload and CDN delivery
Tailwind CSS	Utility-first responsive CSS framework
react-hook-form	Client-side form validation
bcryptjs	Secure password hashing
Vercel	Cloud deployment and hosting

5.2 User Module

The user module handles registration, login, profile management, and session control. Users register with their name, email, and password. Passwords are hashed using bcryptjs before storage in Supabase. Authentication is managed by NextAuth.js using the Credentials provider with JWT-based HTTP-only cookie sessions. Protected pages verify the session server-side before rendering.

5.3 Advertisement Module

The advertisement module allows registered users to create, edit, delete, and manage book listings. When posting an advertisement, users fill in book title, author, ISBN, category, price, condition, description, location (division and district), and upload images. Images are processed through a Next.js API route that uploads them to Cloudinary and returns the stored URLs, which are then saved in the advertisements table of the Supabase database.

5.4 Search and Filter Module

The search and filter module provides book discovery features to all users. The search page displays all active book advertisements in a paginated grid. Users can filter listings by keyword (title or author), category, condition, price range, and location. Filters are applied dynamically using the Supabase query builder with appropriate filtering methods. Results are server-rendered for fast initial load and better SEO indexing.

5.5 Messaging Module

The messaging module enables direct communication between buyers and sellers within the platform. Clicking the Contact Seller button on an advertisement creates a new conversation or opens the existing one between the buyer and seller for that specific listing. The inbox page lists all conversations for the authenticated user. The conversation detail view displays the full message thread with a reply form. Messages are marked as read when the recipient opens the conversation.

5.6 Admin Module

The admin module provides a comprehensive management interface accessible only to users with admin role. The admin dashboard displays summary statistics including total users, active advertisements, pending reports, and total categories. Admins can approve, reject, or delete advertisements; activate or deactivate user accounts; create, edit, or delete categories; and review and resolve user-submitted reports. Role-based access control is enforced through server-side middleware.

5.7 Deployment

The application is deployed on Vercel, which provides automatic CI/CD from the GitHub repository. The Supabase PostgreSQL database and Cloudinary image storage are managed as external cloud services. Environment variables for database connection, authentication secrets, and Cloudinary configuration are stored securely as Vercel project environment variables. The live application is accessible at <https://book-cycle-taupe.vercel.app>.

Chapter 6: User Interface

6.1 Homepage

The homepage is the default landing page of BookCycleBD. It displays featured and recently posted book advertisements, and quick-access category filters. The navigation bar provides links to Search, Post Advertisement, Messages, and the user Profile. The homepage is designed to give visitors an immediate overview of the platform and encourage them to explore or post listings. It is fully responsive and loads quickly using Next.js server-side rendering.

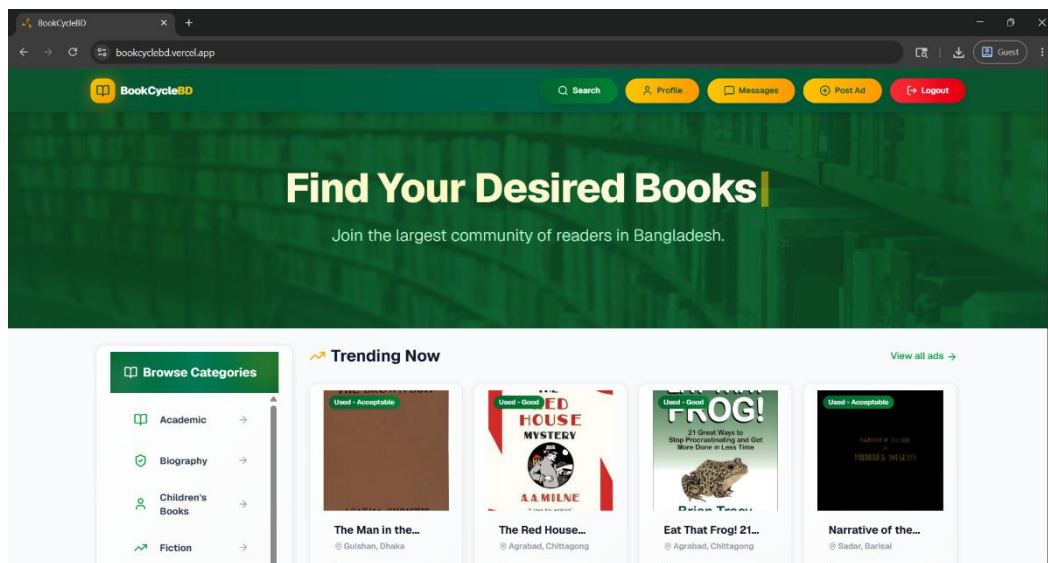


Figure 4.1: Homepage

6.2 Login and Registration

Users must register and log in before posting advertisements or sending messages. The registration page collects name, email, and password. Passwords are validated for minimum length on the client side using react-hook-form and hashed server-side using bcryptjs before storage in Supabase. The login page uses NextAuth.js Credentials provider with a JWT session strategy. Sessions persist across browser tabs via HTTP-only cookies, preventing XSS attacks.

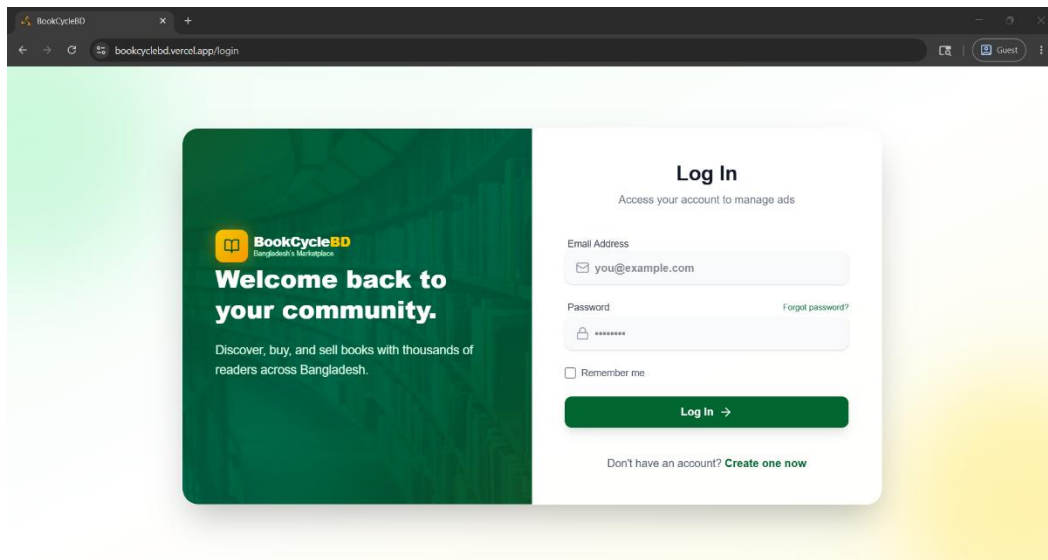


Figure 4.2: Login Page

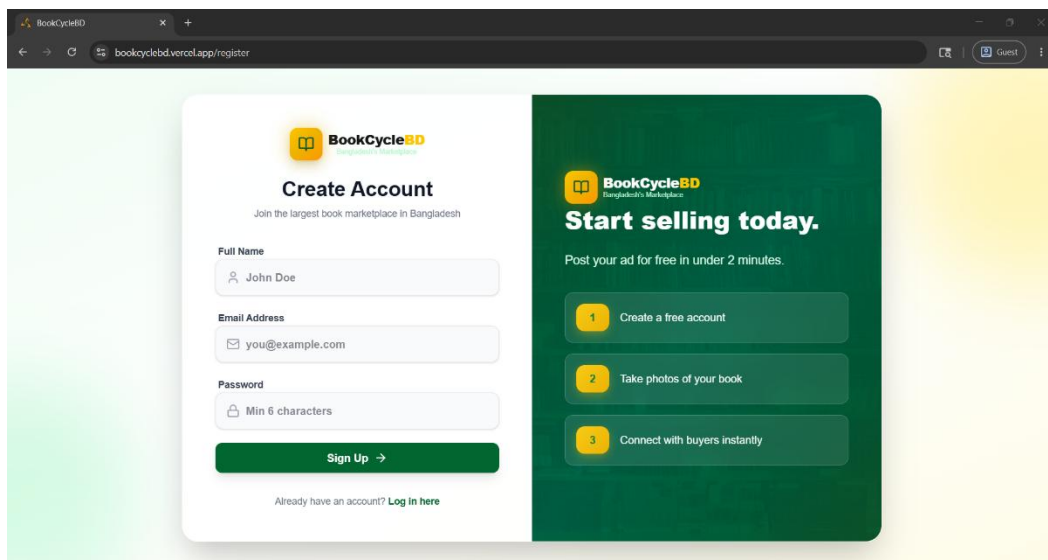


Figure 4.3: Registration Page

6.3 Search Books

The search page displays all active book advertisements in a paginated grid. Users can filter listings by keyword (title or author), category, condition, price range, division, and district. Results are server-rendered for fast initial load and SEO indexing. The search interface provides intuitive controls that allow users to narrow down results quickly and efficiently.

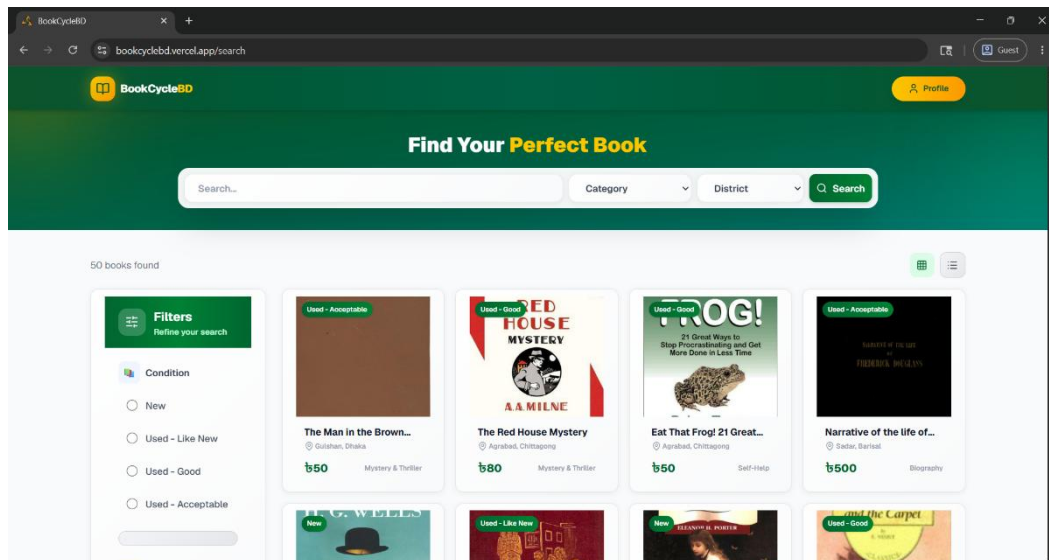


Figure 4.4: Search Books

6.4 Post Advertisement

The Post Advertisement page allows authenticated users to create a new book listing. The form collects book title, author name, ISBN, category, price, condition, description, location, and images. Client-side validation ensures that required fields are not empty before submission. After successful submission, images are uploaded to Cloudinary, and advertisement data is stored in the Supabase PostgreSQL database. This page is accessible only to logged-in users. If an unauthenticated user attempts to access it, the system redirects the user to the login page.

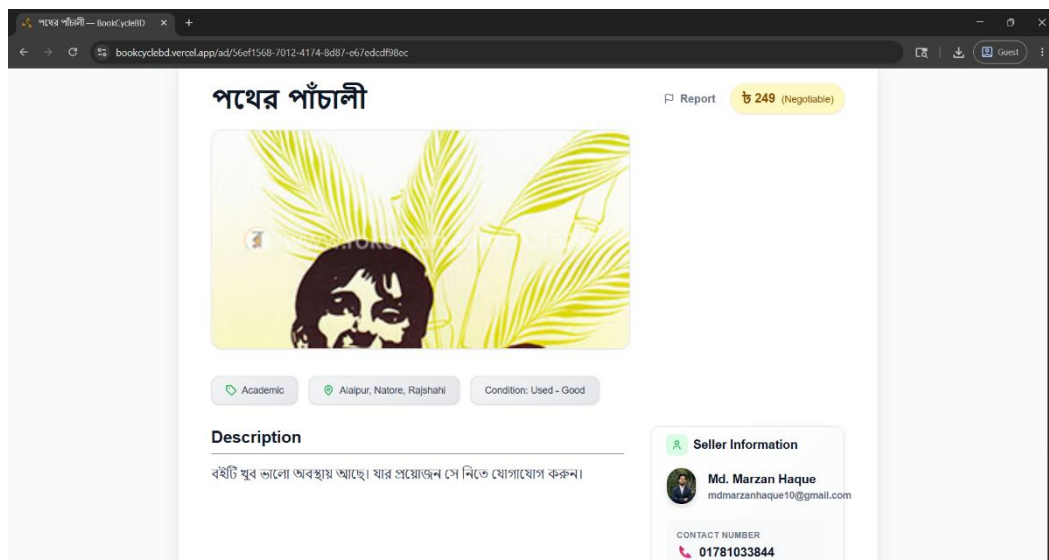


Figure 4.5: Book Advertisement Detail

Sell Your Books

Turn your unread pages into cash. Reach thousands of book lovers across Bangladesh in minutes.

Posting as: mdmarzanhaque10@gmail.com

Step 1 of 4 Details

Details Pricing Photos Review

Ad Title *

e.g. Cambridge IELTS 15 (Like New)

Condition * **Category ***

Select Condition Select a Category

Description *

Describe your book (e.g., condition, edition, author, why you are selling it...)

Figure 4.6: Post Advertisement Form

6.5 Messaging

The messaging system allows buyers and sellers to communicate directly within the platform. Clicking Contact Seller on an advertisement creates a new conversation or opens the existing one between the buyer and seller for that specific advertisement. The inbox page lists all conversations for the authenticated user. The conversation detail view shows the full message thread with a reply form. Messages are marked as read when the recipient opens the conversation.

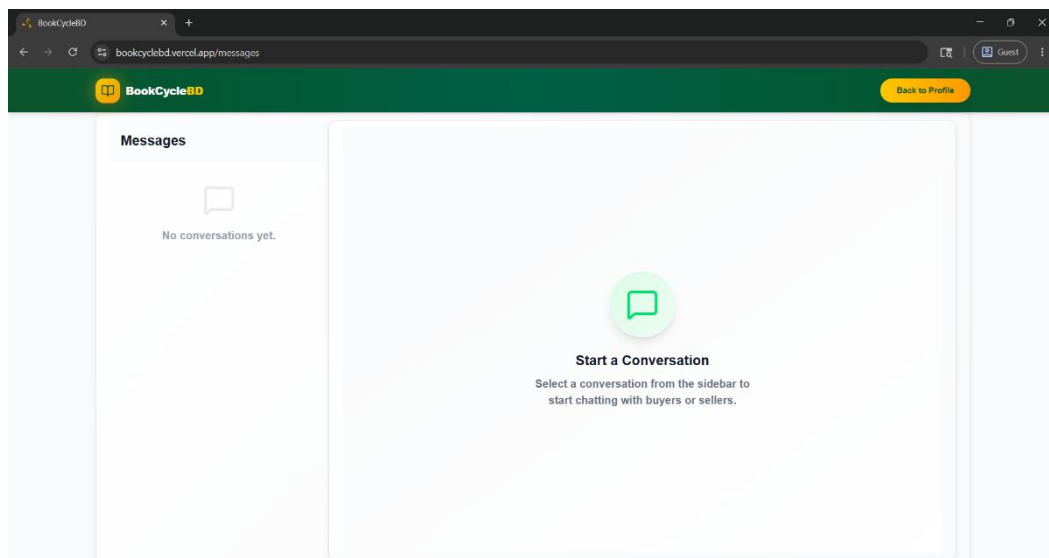


Figure 4.7: Messaging Inbox

6.6 Seller Dashboard

The My Ads section in the user profile displays all advertisements posted by the authenticated user, including their status (active, sold, pending, or rejected) and view count. Sellers can

quickly access the edit or delete options for each listing from this page. Sellers can edit any of their own active listings. The edit form is pre-populated with the current values of the advertisement, allowing updates to title, author, price, condition, description, location, and images.

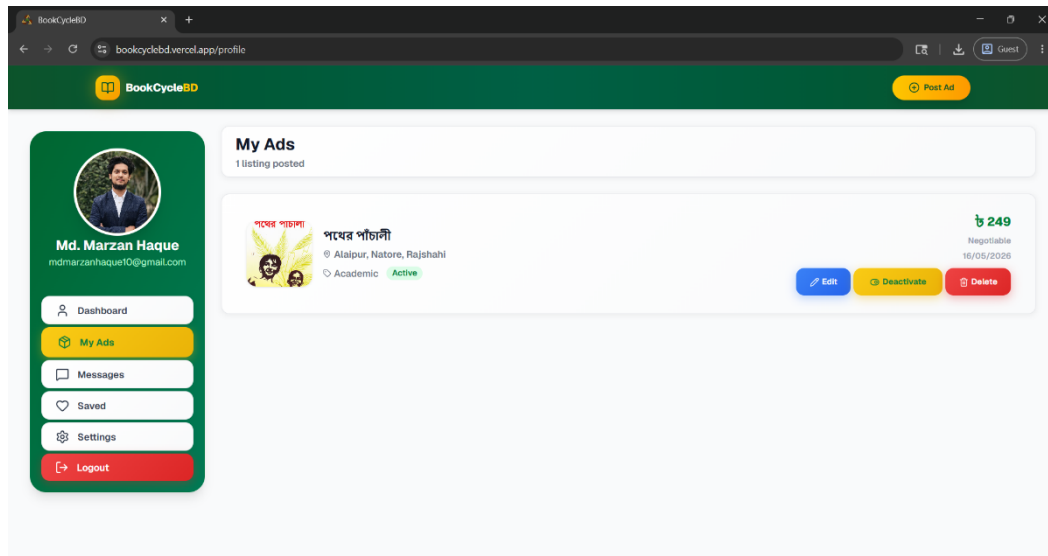


Figure 4.8: My Advertisements Page

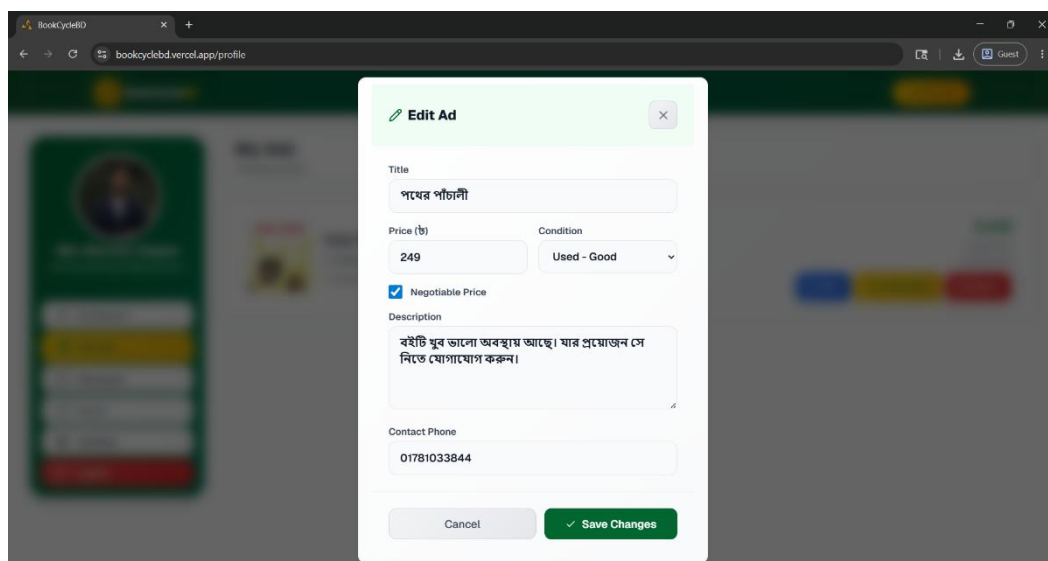


Figure 4.9: Edit Advertisement

6.7 Admin Dashboard

The admin dashboard is accessible only to users with the admin role in the database. It displays summary statistics including total users, active advertisements, pending reports, and total categories. Quick-access navigation links lead to each management section. The admin panel is protected by server-side middleware that verifies the user role on every request.

The user management section displays a table of all registered users with their name, email, registration date, role, and account status. Admins can activate or deactivate user accounts and assign or revoke admin roles. The advertisement management page allows admins to approve, reject, or delete any listing. The category management section provides a full interface for adding, editing, and deleting book categories. The reports management page lists all user-submitted reports, which admins can review and resolve.

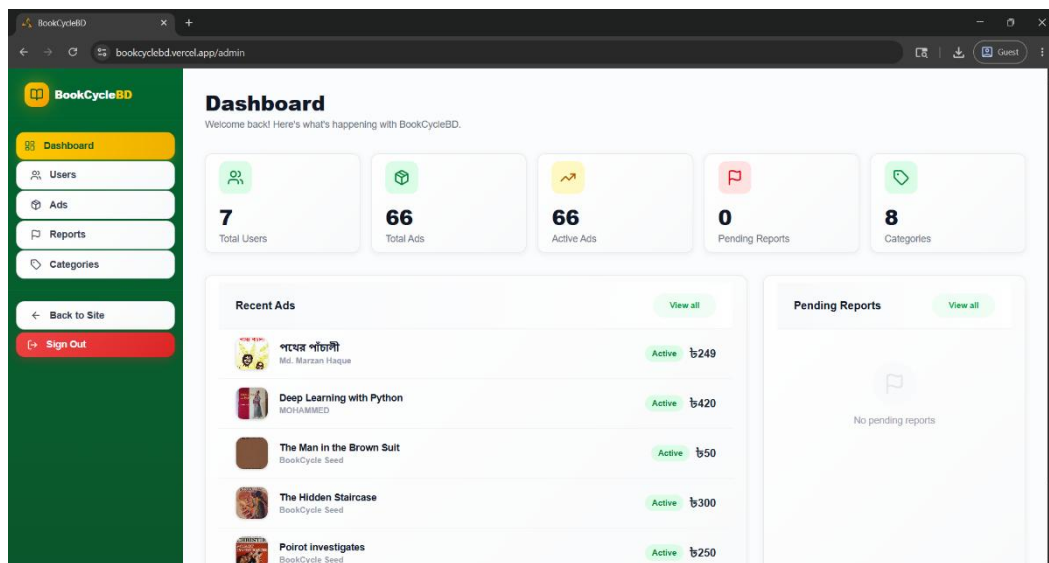


Figure 4.10: Admin Dashboard

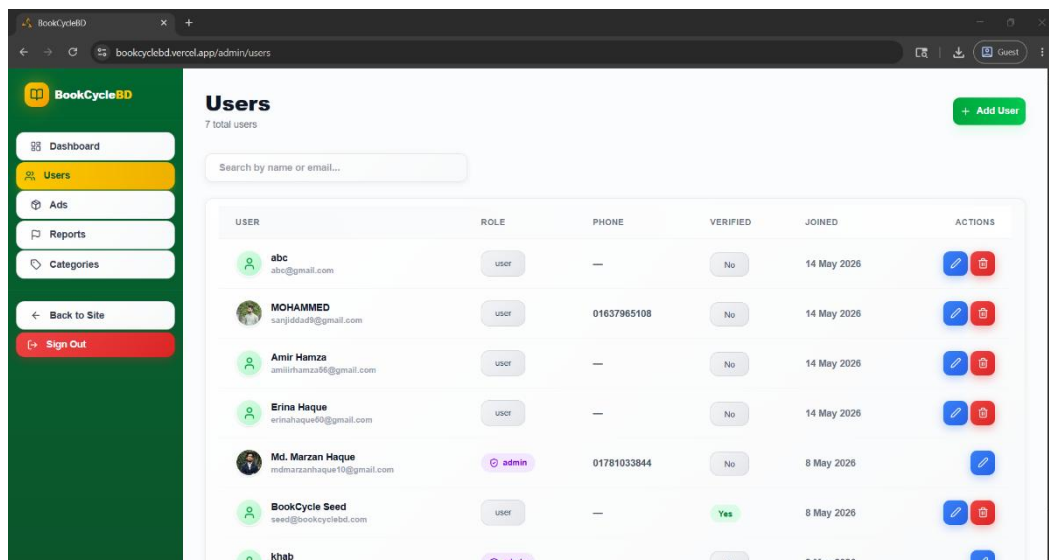


Figure 4.11: Manage Users

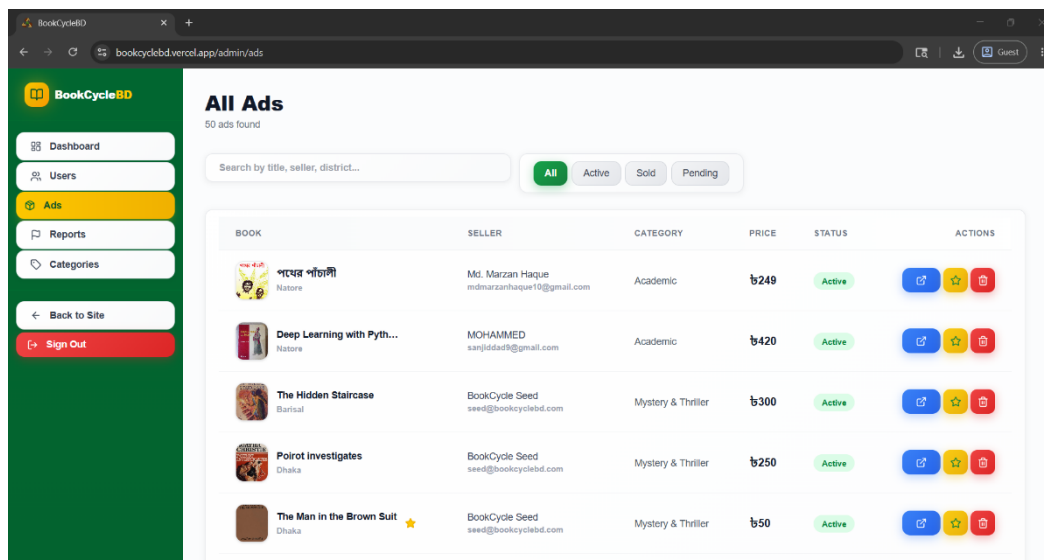


Figure 4.12: Manage Advertisements

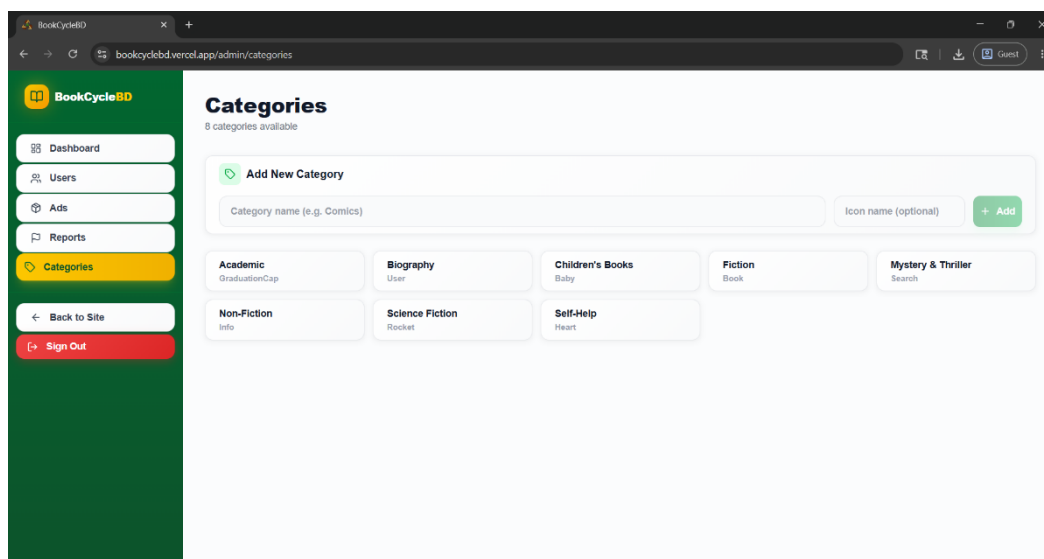


Figure 4.13: Manage Categories

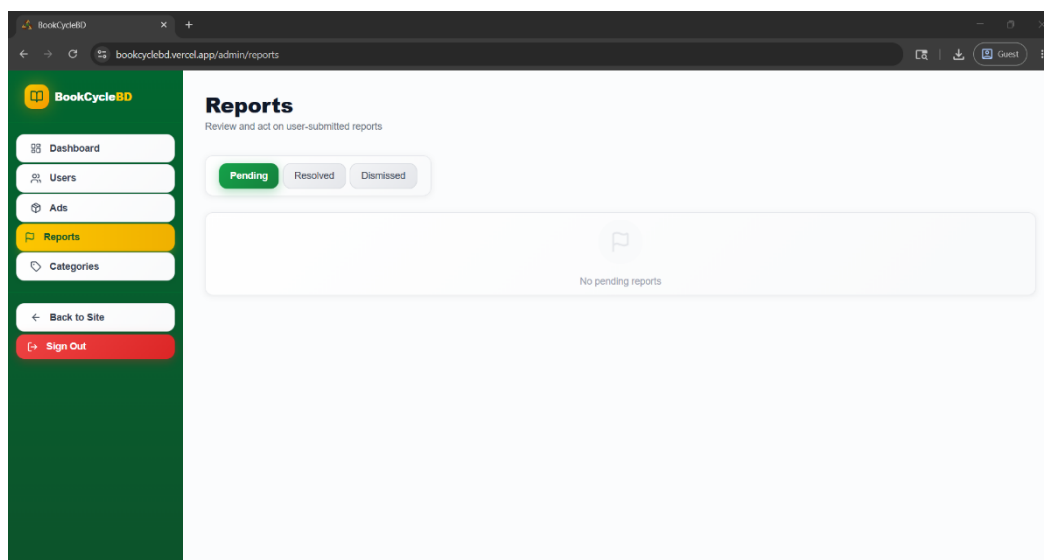


Figure 4.14: Manage Reports

Chapter 7: Testing and Evaluation

7.1 Testing Strategy

Testing was conducted to verify the correctness, usability, and performance of BookCycleBD. Three types of testing were performed: functional testing, usability testing, and performance testing. Functional testing verified that all system features work as expected. Usability testing evaluated the user interface and user experience. Performance testing measured the speed and efficiency of the deployed application.

7.2 Functional Testing

Functional testing was conducted to verify that all features and functionalities of BookCycleBD work as expected. This included testing various scenarios such as user registration, login, posting advertisements, searching and filtering, messaging, and administrative operations. All major functions were tested and verified successfully.

7.3 Usability Testing

Usability testing aimed to evaluate the user interface and overall user experience of BookCycleBD. The interface was assessed for navigation clarity, information layout, and ease of use for buyers, sellers, and administrators. The Tailwind CSS-based interface was found to be clean, modern, and intuitive. The platform was tested on mobile (375px), tablet (768px), and desktop (1280px) viewports, and all pages adapted correctly to different screen sizes. The Lighthouse accessibility score of 95/100 confirms that the platform meets basic WCAG standards, including sufficient color contrast and proper semantic HTML.

7.4 Performance Testing

Performance testing was conducted using Chrome DevTools Lighthouse on the deployed version of BookCycleBD. The test was performed on the production build hosted on Vercel.

Metric	Result	Target
Performance Score	91/100	90/100
First Contentful Paint (FCP)	1.2 seconds	< 1.8 seconds
Largest Contentful Paint (LCP)	1.9 seconds	< 2.5 seconds
Time to Interactive (TTI)	2.1 seconds	< 3.0 seconds
Cumulative Layout Shift (CLS)	0.04	< 0.1
Accessibility Score	95/100	> 90/100

7.5 Test Case Results

The following table presents the complete functional test case results for BookCycleBD:

Test ID	Scenario	Input / Action	Expected Result	Status
TC-01	User registration	Valid name, email, password	Account created successfully	Passed
TC-02	Duplicate registration	Existing email address	Error message displayed	Passed
TC-03	User login	Valid email and password	User logged in successfully	Passed
TC-04	Invalid login	Wrong password	Login failed with error message	Passed
TC-05	Post advertisement	Valid book details and image	Advertisement created successfully	Passed
TC-06	Empty advertisement form	Submit without required fields	Validation message displayed	Passed
TC-07	Search by title	Enter book title keyword	Matching books displayed	Passed
TC-08	Filter by category	Select category	Books from selected category displayed	Passed
TC-09	Filter by price	Enter min and max price	Books within price range displayed	Passed
TC-10	Contact seller	Click Contact Seller button	Conversation created or opened	Passed
TC-11	Send message	Enter message and submit	Message saved and displayed	Passed
TC-12	Edit own advertisement	Update listing details	Advertisement updated	Passed
TC-13	Delete own advertisement	Click delete option	Advertisement removed	Passed
TC-14	Unauthorized edit attempt	Try to edit another user ad	Access denied	Passed
TC-15	Admin login	Login as admin	Admin dashboard accessible	Passed
TC-16	Non-admin admin access	Login as normal user	Admin dashboard inaccessible	Passed
TC-17	Manage users	Admin changes user status	User status updated	Passed
TC-18	Manage advertisements	Admin approves or rejects ad	Advertisement status updated	Passed
TC-19	Manage categories	Admin adds or edits category	Category saved successfully	Passed
TC-20	Report advertisement	User submits report	Report stored in database	Passed
TC-21	Responsive design	Open on mobile, tablet, desktop	Layout adapts correctly	Passed
TC-22	Logout	Click logout	Session ended successfully	Passed

Chapter 8: Conclusion and Future Scope

8.1 Conclusion

BookCycleBD is a web-based second-hand book marketplace developed to support students and readers in Bangladesh by providing an organized platform for buying and selling used books. The system allows users to register, post advertisements, upload images, search and filter books, communicate with sellers, and manage their own listings. It also includes an admin panel for managing users, advertisements, categories, and reports.

The project demonstrates the application of modern full-stack web development technologies, database design, authentication, image management, responsive user interface design, testing, and cloud deployment. Through functional, usability, and performance testing, the system was found to meet its major objectives. Overall, BookCycleBD provides a practical and scalable foundation for a dedicated second-hand book marketplace and can be further improved with payment, delivery, rating, and mobile application features.

8.2 Limitations

Although BookCycleBD successfully implements the core features of a second-hand book marketplace, it has some limitations. The current version does not support online payment, delivery tracking, seller rating, real-time chat, mobile applications, or automated fraud detection. Transactions are completed through direct communication between buyers and sellers. These limitations can be addressed in future versions of the system.

8.3 Future Scope

BookCycleBD lays a strong foundation for future enhancements and expansions. Some potential areas for future development include:

- **Payment Gateway Integration:** Integrate bKash or SSLCommerz to enable secure in-platform payments so buyers and sellers can complete transactions without meeting in person.
- **Real-time Messaging:** Implement Supabase Realtime subscriptions so messages appear instantly without page refresh, providing a true chat-like experience.
- **Mobile Application:** Develop dedicated Android and iOS applications using React Native to reach the large portion of Bangladeshi users who primarily use smartphones.
- **Email Notifications:** Send email alerts for new messages, advertisement approval status, and account activity using services such as Resend or SendGrid.
- **Ratings and Reviews:** Allow buyers to rate and review sellers after a transaction, building a trust and reputation system within the community.
- **Multi-language Support:** Add Bengali language support to make the platform accessible to a wider audience across Bangladesh.

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